2024

STATISTICS — HONOURS

Paper : DSCC-1

(Descriptive Statistics - I and Probability - I)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** questions : 2×5

(a) Define a nominal variable. Give one example.

(b) Define percentile of order 25 of a set of n observations of a variable x.

(c) If $y = 3 - \frac{x}{2}$, and the A.M. and variance of x are both 4, find those of y.

(d) Give one real life example, where the mode is a suitable measure of central tendency.

(e) What do you mean by a symmetric frequency distribution?

(f) What is a random experiment?

(g) State Bayes' theorem.

(h) If three are selected from a group of 4 boys and 4 girls, what is the probability that at least one is a boy and at least one is a girl?

2. Answer ***any four*** questions :

(a) Suppose we have $k(\geq 2)$ groups, the i-th group having n_i observations with arithmetic mean $\bar{x}_i$ and standard deviation s_i, $i = 1, 2, ..., k$. Find an expression for the standard deviation of the combined group. 5

(b) If $x_1, x_2, ..., x_n$ are n observations of a variable x with range R and standard deviation s, show that $s^2 \geq \frac{R^2}{2n}$. Discuss the case of equality other than when all x_i's are constant. 3+2

(c) Define lower and upper quartiles. What is the measure of skewness based on quartiles? Find out its limits. (1+1)+1+2

(d) Define conditional probability. Show that it satisfies all the axioms of probability. 2+3

(e) Two people toss a fair coin n times each. Find the probability that they will score the same number of tails. 5

(f) Define mutually exclusive and independent events. Discuss whether any one of them implies the other or not. 2+3

3. Answer ***any three*** questions :

(a) Define geometric and harmonic mean of a set of positive observations. Show that the latter cannot exceed the former. When will these two be equal? Give one useful example of each of these two measures. (2+2)+6+1+(2+2)

(b) (i) Suppose for a frequency distribution with k classes, the upper class boundary bears a constant ratio r to the lower class boundary. If x_i and f_i be respectively the class mark and frequency of the i-th class and G be the geometric mean of the distribution, then show that

$$\ln G = \ln x_1 + \frac{\ln r}{N}\sum_{i=1}^{k}(i-1)f_i, \quad N = \sum_{i=1}^{k} f_i.$$

(ii) The mean and standard deviation of n observations $x_1, x_2, \ldots, x_n$ were found to be A and s, respectively. It was later discovered that one of the observations was wrongly noted as x_j instead of its original value $x'_j = x_j + nh$. Show that the corrected mean and standard deviation are respectively given by

$$A^* = A + h,$$
$$s^* = \sqrt{s^2 + (n-1)h^2 + 2h(x_j - A)}.$$

(iii) Stating all necessary results, prove that the difference between the mean and the median cannot exceed the standard deviation. When does the equality hold? 5+5+5

(c) (i) What are raw and central moments of a set of observations related to some variable? Express r-th order central moment in terms of r-th and lower order raw moments.

(ii) State and prove Poincare's theorem for any n events $A_1, A_2, \ldots, A_n$. (3+4)+8

(d) (i) Define and illustrate the following terms through an example : Trial, sample space, sample points, elementary and composite events.

(ii) If A, B, C and D are independent events, show that $A \cup B$ and $C \cup D$ are also independent. 10+5

(e) (i) State and prove the theorem of compound probability.

(ii) Let $\{B_i\}$ be a sequence of exhaustive and mutually exclusive events. If A is another event, then for any other event C, show that

$$P(C \mid A) = \sum_i P(B_i)P(A \mid B_i)P(C \mid A \cap B_i) \Big/ \sum_i P(B_i)P(A \mid B_i).$$

(iii) Among the 270 tickets sold in a lottery, 100 are coloured red, 90 coloured blue and 80 coloured green. What are the probabilities that (I) blue tickets will win both 1st and 2nd prizes; (II) a blue ticket will win the 1st prize and a green ticket the 2nd? 4+6+5